

Problem 15: Quotient Rule with Radicals (Square Roots)

Course: Mathematics-I (Differential Calculus)

Textbook Page: 31 | Problem: 15

প্রদত্ত সমীকরণ: $y = \frac{\sqrt{x} + 1}{\sqrt{x} - 1}$

নির্ণেয়: $\frac{dy}{dx}$

ENGLISH NOTE — ALGEBRAIC RADICAL QUOTIENT FORMULATION

The given function is a rational function with square root radicals:

$$y = \frac{u(x)}{v(x)}, \quad \text{where } u(x) = \sqrt{x} + 1, \quad v(x) = \sqrt{x} - 1$$

Applying the standard Quotient Rule:

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}, \quad \text{with } \frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

উপাদান (Component)	ফাংশন রূপ	প্রমিত অন্তরজ সূত্র (Standard Derivative)
লব (u)	$\sqrt{x} + 1$	$\frac{d}{dx}(\sqrt{x} + 1) = \frac{1}{2\sqrt{x}}$
হর (v)	$\sqrt{x} - 1$	$\frac{d}{dx}(\sqrt{x} - 1) = \frac{1}{2\sqrt{x}}$
হরের বর্গ (v^2)	$(\sqrt{x} - 1)^2$	$(\sqrt{x} - 1)^2$ (উৎপাদক আকারেই রাখা সমীচীন)

ENGLISH NOTE — CRITICAL EXAM PITFALLS & ALGEBRAIC TRAPS

1. Sign Distribution in Numerator: When factoring out $\frac{1}{2\sqrt{x}}$, preserve parentheses:

$$(\sqrt{x} - 1) - (\sqrt{x} + 1) = \sqrt{x} - 1 - \sqrt{x} - 1 = -2$$

2. Fractional Level Placement: Notice that $\frac{1}{2\sqrt{x}} \times (-2) = -\frac{1}{\sqrt{x}}$. Bring $\sqrt{x}$ into the main denominator instead of leaving a complex multi-deck fraction.

২. ধাপে ধাপে পূর্ণাঙ্গ সমাধান (Step-by-Step Solution)

ধাপ ১: উভয় পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করে:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{\sqrt{x} + 1}{\sqrt{x} - 1} \right)$$

ধাপ ২: ভাগফলের সূত্র (Quotient Rule) অনুসারে সাজানো:

$$\frac{dy}{dx} = \frac{(\sqrt{x} - 1) \frac{d}{dx}(\sqrt{x} + 1) - (\sqrt{x} + 1) \frac{d}{dx}(\sqrt{x} - 1)}{(\sqrt{x} - 1)^2}$$

ধাপ ৩: অন্তরজের মান $\frac{d}{dx}(\sqrt{x} \pm 1) = \frac{1}{2\sqrt{x}}$ বসিয়ে:

$$\frac{dy}{dx} = \frac{(\sqrt{x} - 1) \cdot \frac{1}{2\sqrt{x}} - (\sqrt{x} + 1) \cdot \frac{1}{2\sqrt{x}}}{(\sqrt{x} - 1)^2}$$

ধাপ ৪: লব থেকে সাধারণ উৎপাদক $\frac{1}{2\sqrt{x}}$ কমন নিয়ে সরলীকরণ:

$$\frac{dy}{dx} = \frac{\frac{1}{2\sqrt{x}} [(\sqrt{x} - 1) - (\sqrt{x} + 1)]}{(\sqrt{x} - 1)^2} = \frac{\frac{1}{2\sqrt{x}} [\sqrt{x} - 1 - \sqrt{x} - 1]}{(\sqrt{x} - 1)^2} = \frac{\frac{1}{2\sqrt{x}} \cdot (-2)}{(\sqrt{x} - 1)^2}$$

ধাপ ৫: লব ও হরের ভগ্নাংশ সরল করে চূড়ান্ত প্রমিত আকারে প্রকাশ:

$$\frac{dy}{dx} = \frac{-\frac{1}{\sqrt{x}}}{(\sqrt{x} - 1)^2} = -\frac{1}{\sqrt{x}(\sqrt{x} - 1)^2}$$

চূড়ান্ত উত্তর (FINAL STANDARD ANSWER)

$$\frac{dy}{dx} = -\frac{1}{\sqrt{x}(\sqrt{x} - 1)^2}$$



ENGLISH NOTE — DOMAIN, SINGULARITY & VERTICAL TANGENT

- **Domain of $y(x)$:** Radical requires $x \geq 0$, and non-zero denominator requires $\sqrt{x} \neq 1 \implies x \neq 1$. Thus, $\text{Dom}(y) = [0, 1) \cup (1, \infty)$.
- **Domain of Differentiability ($\frac{dy}{dx}$):** Because $\frac{1}{\sqrt{x}}$ causes division by zero at $x = 0$, the derivative exhibits a vertical asymptote ($\frac{dy}{dx} \rightarrow -\infty$). Thus, $\text{Dom}\left(\frac{dy}{dx}\right) = (0, 1) \cup (1, \infty)$.



ENGLISH NOTE — RAPID VERIFICATION VIA SUBSTITUTION & CHAIN RULE

Let $u = \sqrt{x}$. Then $y = \frac{u+1}{u-1} = 1 + \frac{2}{u-1} = 1 + 2(u-1)^{-1}$.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = [-2(u-1)^{-2}] \cdot \left(\frac{1}{2\sqrt{x}}\right) = -\frac{1}{\sqrt{x}(u-1)^2} = -\frac{1}{\sqrt{x}(\sqrt{x}-1)^2}$$

This provides an independent and lightning-fast sanity check in engineering exams!

🎯 AMIE EXAM MARKS DISTRIBUTION STRATEGY

Exam Evaluation Step	Required Mathematical Rigor	Marks
Quotient Rule Formulation	Correct expansion of $\frac{vu' - uv'}{v^2}$	1.5
Radical Differentiation	Accurate derivative of $\sqrt{x} \implies \frac{1}{2\sqrt{x}}$	1.5
Algebraic Simplification	Factoring $\frac{1}{2\sqrt{x}}$ and simplifying -2	1.0
Final Standard Format	Non-nested fraction $-\frac{1}{\sqrt{x}(\sqrt{x}-1)^2}$	1.0
Total AMIE Section-A Marks		5.0 / 5.0